PGCP-AC August 26
Database Assignment No-3

1. Write a query to calculate the total salary of all employees.

```
mysql> select sum(salary) as total_salary from employee;
+--------------+
| total_salary |
+--------------+
|   1132000.00 |
+--------------+
1 row in set (0.00 sec)
```

2. Write a query to find the average salary of employees in each department using GROUP BY.

```
mysql> select dept_id, avg(salary) as Average_salary from employee
    -> group by dept_id;
+---------+----------------+
| dept_id | Average_salary |
+---------+----------------+
|      10 |   60000.000000 |
|      20 |   49200.000000 |
|      30 |   63333.333333 |
|      40 |   48666.666667 |
+---------+----------------+
4 rows in set (0.00 sec)
```

3. Write a query to count the total number of employees in each department.

```
mysql> select dept_id,count(*) as total_employees from employee
    -> group by dept_id;
+---------+-----------------+
| dept_id | total_employees |
+---------+-----------------+
|      10 |               7 |
|      20 |               6 |
|      30 |               6 |
|      40 |               4 |
+---------+-----------------+
4 rows in set (0.00 sec)
```

4. Write a query to display departments having more than 5 employees using HAVING clause.

```
mysql> select dept_id, count(dept_id) as total_employees from employee
    -> group by dept_id
    -> having count(dept_id) > 5;
+---------+-----------------+
| dept_id | total_employees |
+---------+-----------------+
|      10 |               7 |
|      20 |               6 |
|      30 |               6 |
+---------+-----------------+
3 rows in set (0.00 sec)
```

5. Write a query to list distinct department locations from the Dept table.

```
mysql> select distinct city from department;
+--------+
| city   |
+--------+
| Mumbai |
| Pune   |
| Delhi  |
| Nagpur |
+--------+
4 rows in set (0.00 sec)
```

6. Write a query to display the highest salary among all employees. Write a query to display all total number of employee in table.

```
mysql> select max(salary) as highest_salary from employee;
+----------------+
| highest_salary |
+----------------+
|       85000.00 |
+----------------+
1 row in set (0.01 sec)
```

```
mysql> select count(emp_id) as total_Employee from employee;
+----------------+
| total_Employee |
+----------------+
|             23 |
+----------------+
1 row in set (0.00 sec)
```

7. Write a query to find employees whose name starts with 'A' using LIKE operator.

```
mysql> select * from employee where name like 'A%';
+--------+--------+----------+---------+--------------+
| emp_id | name   | salary   | dept_id | joining_date |
+--------+--------+----------+---------+--------------+
|    101 | Aarav  | 75000.00 |      10 | 2020-03-15   |
|    103 | Arun   | 58000.00 |      10 | 2019-11-05   |
|    105 | Adrian | 52000.00 |      10 | 2021-12-31   |
|    108 | Aman   | 39000.00 |      20 | 2020-07-07   |
|    112 | Aarhan | 61000.00 |      20 | 2020-01-01   |
|    114 | Arav   | 85000.00 |      30 | 2019-04-22   |
|    119 | Asha   | 53000.00 |      30 | 2020-11-11   |
|    121 | Anand  | 41000.00 |      40 | 2021-09-01   |
+--------+--------+----------+---------+--------------+
8 rows in set (0.00 sec)
```

8. Write a query to find employees whose name ends with 'n' using LIKE operator.

```
mysql> select * from employee where name like '%n';
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    102 | Karan    | 62000.00 |      10 | 2021-06-20   |
|    103 | Arun     | 58000.00 |      10 | 2019-11-05   |
|    105 | Adrian   | 52000.00 |      10 | 2021-12-31   |
|    106 | Varun    | 68000.00 |      10 | 2020-08-12   |
|    108 | Aman     | 39000.00 |      20 | 2020-07-07   |
|    109 | Rohan    | 42000.00 |      20 | 2021-02-18   |
|    112 | Aarhan   | 61000.00 |      20 | 2020-01-01   |
|    115 | Kiran    | 72000.00 |      30 | 2020-12-12   |
|    116 | Mohan    | 64000.00 |      30 | 2021-07-30   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
|    123 | Brian    |     NULL |      40 | 2023-08-08   |
+--------+----------+----------+---------+--------------+
11 rows in set (0.00 sec)
```

9. Write a query to find employees whose name contains 'ra' using LIKE operator.

```
mysql> select * from employee where name like '%ra%';
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    101 | Aarav    | 75000.00 |      10 | 2020-03-15   |
|    102 | Karan    | 62000.00 |      10 | 2021-06-20   |
|    104 | Mira     | 45000.00 |      10 | 2022-01-10   |
|    110 | Pranav   | 48000.00 |      20 | 2018-09-09   |
|    111 | Meera    | 56000.00 |      20 | 2021-10-25   |
|    113 | Ira      |     NULL |      20 | 2024-05-12   |
|    114 | Arav     | 85000.00 |      30 | 2019-04-22   |
|    115 | Kiran    | 72000.00 |      30 | 2020-12-12   |
|    117 | Sarah    | 59000.00 |      30 | 2022-06-14   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
+--------+----------+----------+---------+--------------+
10 rows in set (0.00 sec)
```

10. Write a query to display all employees sorted by their Salary in descending order.

```
mysql> select * from employee order by salary desc;
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    114 | Arav     | 85000.00 |      30 | 2019-04-22   |
|    101 | Aarav    | 75000.00 |      10 | 2020-03-15   |
|    115 | Kiran    | 72000.00 |      30 | 2020-12-12   |
|    106 | Varun    | 68000.00 |      10 | 2020-08-12   |
|    122 | Rita     | 67000.00 |      40 | 2020-02-29   |
|    116 | Mohan    | 64000.00 |      30 | 2021-07-30   |
|    102 | Karan    | 62000.00 |      10 | 2021-06-20   |
|    112 | Aarhan   | 61000.00 |      20 | 2020-01-01   |
|    117 | Sarah    | 59000.00 |      30 | 2022-06-14   |
|    103 | Arun     | 58000.00 |      10 | 2019-11-05   |
|    111 | Meera    | 56000.00 |      20 | 2021-10-25   |
|    119 | Asha     | 53000.00 |      30 | 2020-11-11   |
|    105 | Adrian   | 52000.00 |      10 | 2021-12-31   |
|    110 | Pranav   | 48000.00 |      20 | 2018-09-09   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
|    104 | Mira     | 45000.00 |      10 | 2022-01-10   |
|    109 | Rohan    | 42000.00 |      20 | 2021-02-18   |
|    121 | Anand    | 41000.00 |      40 | 2021-09-01   |
|    108 | Aman     | 39000.00 |      20 | 2020-07-07   |
|    120 | Dev      | 38000.00 |      40 | 2017-05-19   |
|    107 | Nina     |     NULL |      10 | 2023-04-01   |
|    113 | Ira      |     NULL |      20 | 2024-05-12   |
|    123 | Brian    |     NULL |      40 | 2023-08-08   |
+--------+----------+----------+---------+--------------+
23 rows in set (0.00 sec)
```

11. Write a query to display all employees sorted by DeptID ascending and then Salary descending.

```
mysql> select * from employee order by emp_id asc, salary desc;
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    101 | Aarav    | 75000.00 |      10 | 2020-03-15   |
|    102 | Karan    | 62000.00 |      10 | 2021-06-20   |
|    103 | Arun     | 58000.00 |      10 | 2019-11-05   |
|    104 | Mira     | 45000.00 |      10 | 2022-01-10   |
|    105 | Adrian   | 52000.00 |      10 | 2021-12-31   |
|    106 | Varun    | 68000.00 |      10 | 2020-08-12   |
|    107 | Nina     |     NULL |      10 | 2023-04-01   |
|    108 | Aman     | 39000.00 |      20 | 2020-07-07   |
|    109 | Rohan    | 42000.00 |      20 | 2021-02-18   |
|    110 | Pranav   | 48000.00 |      20 | 2018-09-09   |
|    111 | Meera    | 56000.00 |      20 | 2021-10-25   |
|    112 | Aarhan   | 61000.00 |      20 | 2020-01-01   |
|    113 | Ira      |     NULL |      20 | 2024-05-12   |
|    114 | Arav     | 85000.00 |      30 | 2019-04-22   |
|    115 | Kiran    | 72000.00 |      30 | 2020-12-12   |
|    116 | Mohan    | 64000.00 |      30 | 2021-07-30   |
|    117 | Sarah    | 59000.00 |      30 | 2022-06-14   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
|    119 | Asha     | 53000.00 |      30 | 2020-11-11   |
|    120 | Dev      | 38000.00 |      40 | 2017-05-19   |
|    121 | Anand    | 41000.00 |      40 | 2021-09-01   |
|    122 | Rita     | 67000.00 |      40 | 2020-02-29   |
|    123 | Brian    |     NULL |      40 | 2023-08-08   |
+--------+----------+----------+---------+--------------+
23 rows in set (0.00 sec)
```

12. Write a query to find employees whose salary is between 30,000 and 60,000.

```
mysql> select * from employee where salary between 30000 and 60000;
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    103 | Arun     | 58000.00 |      10 | 2019-11-05   |
|    104 | Mira     | 45000.00 |      10 | 2022-01-10   |
|    105 | Adrian   | 52000.00 |      10 | 2021-12-31   |
|    108 | Aman     | 39000.00 |      20 | 2020-07-07   |
|    109 | Rohan    | 42000.00 |      20 | 2021-02-18   |
|    110 | Pranav   | 48000.00 |      20 | 2018-09-09   |
|    111 | Meera    | 56000.00 |      20 | 2021-10-25   |
|    117 | Sarah    | 59000.00 |      30 | 2022-06-14   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
|    119 | Asha     | 53000.00 |      30 | 2020-11-11   |
|    120 | Dev      | 38000.00 |      40 | 2017-05-19   |
|    121 | Anand    | 41000.00 |      40 | 2021-09-01   |
+--------+----------+----------+---------+--------------+
12 rows in set (0.01 sec)
```

```
mysql> select * from employee where salary <60000 and salary >30000;
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    103 | Arun     | 58000.00 |      10 | 2019-11-05   |
|    104 | Mira     | 45000.00 |      10 | 2022-01-10   |
|    105 | Adrian   | 52000.00 |      10 | 2021-12-31   |
|    108 | Aman     | 39000.00 |      20 | 2020-07-07   |
|    109 | Rohan    | 42000.00 |      20 | 2021-02-18   |
|    110 | Pranav   | 48000.00 |      20 | 2018-09-09   |
|    111 | Meera    | 56000.00 |      20 | 2021-10-25   |
|    117 | Sarah    | 59000.00 |      30 | 2022-06-14   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
|    119 | Asha     | 53000.00 |      30 | 2020-11-11   |
|    120 | Dev      | 38000.00 |      40 | 2017-05-19   |
|    121 | Anand    | 41000.00 |      40 | 2021-09-01   |
+--------+----------+----------+---------+--------------+
12 rows in set (0.00 sec)
```

13. Write a query to display all employees whose DeptID is in (10, 20, 30).

```
mysql> select * from employee where dept_id between 10 and 30;
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    101 | Aarav    | 75000.00 |      10 | 2020-03-15   |
|    102 | Karan    | 62000.00 |      10 | 2021-06-20   |
|    103 | Arun     | 58000.00 |      10 | 2019-11-05   |
|    104 | Mira     | 45000.00 |      10 | 2022-01-10   |
|    105 | Adrian   | 52000.00 |      10 | 2021-12-31   |
|    106 | Varun    | 68000.00 |      10 | 2020-08-12   |
|    107 | Nina     |     NULL |      10 | 2023-04-01   |
|    108 | Aman     | 39000.00 |      20 | 2020-07-07   |
|    109 | Rohan    | 42000.00 |      20 | 2021-02-18   |
|    110 | Pranav   | 48000.00 |      20 | 2018-09-09   |
|    111 | Meera    | 56000.00 |      20 | 2021-10-25   |
|    112 | Aarhan   | 61000.00 |      20 | 2020-01-01   |
|    113 | Ira      |     NULL |      20 | 2024-05-12   |
|    114 | Arav     | 85000.00 |      30 | 2019-04-22   |
|    115 | Kiran    | 72000.00 |      30 | 2020-12-12   |
|    116 | Mohan    | 64000.00 |      30 | 2021-07-30   |
|    117 | Sarah    | 59000.00 |      30 | 2022-06-14   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
|    119 | Asha     | 53000.00 |      30 | 2020-11-11   |
+--------+----------+----------+---------+--------------+
19 rows in set (0.00 sec)
```

14. Write a query to display Min salary of employee .

```
mysql> select min(salary) from employee;
+-------------+
| min(salary) |
+-------------+
|    38000.00 |
+-------------+
1 row in set (0.00 sec)
```

15. Write a query to display employees whose JoiningDate is between '2020-01-01' and '2021-12-31'.

```
mysql> select * from employee where joining_date between '2020-01-01' and '2021-12-31';
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    101 | Aarav    | 75000.00 |      10 | 2020-03-15   |
|    102 | Karan    | 62000.00 |      10 | 2021-06-20   |
|    105 | Adrian   | 52000.00 |      10 | 2021-12-31   |
|    106 | Varun    | 68000.00 |      10 | 2020-08-12   |
|    108 | Aman     | 39000.00 |      20 | 2020-07-07   |
|    109 | Rohan    | 42000.00 |      20 | 2021-02-18   |
|    111 | Meera    | 56000.00 |      20 | 2021-10-25   |
|    112 | Aarhan   | 61000.00 |      20 | 2020-01-01   |
|    115 | Kiran    | 72000.00 |      30 | 2020-12-12   |
|    116 | Mohan    | 64000.00 |      30 | 2021-07-30   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
|    119 | Asha     | 53000.00 |      30 | 2020-11-11   |
|    121 | Anand    | 41000.00 |      40 | 2021-09-01   |
|    122 | Rita     | 67000.00 |      40 | 2020-02-29   |
+--------+----------+----------+---------+--------------+
14 rows in set (0.00 sec)
```

16. Write a query to display employees whose Salary is NULL.

```
mysql> select  * from employee where salary is null;
+--------+-------+--------+---------+--------------+
| emp_id | name  | salary | dept_id | joining_date |
+--------+-------+--------+---------+--------------+
|    107 | Nina  |   NULL |      10 | 2023-04-01   |
|    113 | Ira   |   NULL |      20 | 2024-05-12   |
|    123 | Brian |   NULL |      40 | 2023-08-08   |
+--------+-------+--------+---------+--------------+
3 rows in set (0.00 sec)
```

17. Write a query to display employees whose Salary is NOT NULL.

```
mysql> select  * from employee where salary is not null;
+--------+----------+----------+---------+--------------+
| emp_id | name     | salary   | dept_id | joining_date |
+--------+----------+----------+---------+--------------+
|    101 | Aarav    | 75000.00 |      10 | 2020-03-15   |
|    102 | Karan    | 62000.00 |      10 | 2021-06-20   |
|    103 | Arun     | 58000.00 |      10 | 2019-11-05   |
|    104 | Mira     | 45000.00 |      10 | 2022-01-10   |
|    105 | Adrian   | 52000.00 |      10 | 2021-12-31   |
|    106 | Varun    | 68000.00 |      10 | 2020-08-12   |
|    108 | Aman     | 39000.00 |      20 | 2020-07-07   |
|    109 | Rohan    | 42000.00 |      20 | 2021-02-18   |
|    110 | Pranav   | 48000.00 |      20 | 2018-09-09   |
|    111 | Meera    | 56000.00 |      20 | 2021-10-25   |
|    112 | Aarhan   | 61000.00 |      20 | 2020-01-01   |
|    114 | Arav     | 85000.00 |      30 | 2019-04-22   |
|    115 | Kiran    | 72000.00 |      30 | 2020-12-12   |
|    116 | Mohan    | 64000.00 |      30 | 2021-07-30   |
|    117 | Sarah    | 59000.00 |      30 | 2022-06-14   |
|    118 | Niranjan | 47000.00 |      30 | 2021-03-03   |
|    119 | Asha     | 53000.00 |      30 | 2020-11-11   |
|    120 | Dev      | 38000.00 |      40 | 2017-05-19   |
|    121 | Anand    | 41000.00 |      40 | 2021-09-01   |
|    122 | Rita     | 67000.00 |      40 | 2020-02-29   |
+--------+----------+----------+---------+--------------+
20 rows in set (0.00 sec)
```

18. Write a query to calculate the total salary per department, but only for departments where total salary is greater than 1,00,000 (use HAVING).

```
mysql> select dept_id, sum(salary) as total_salary from employee
    -> group by dept_id
    -> having total_salary > 100000;
+---------+--------------+
| dept_id | total_salary |
+---------+--------------+
|      10 |    360000.00 |
|      20 |    246000.00 |
|      30 |    380000.00 |
|      40 |    146000.00 |
+---------+--------------+
4 rows in set (0.00 sec)
```

19. Write a query to display all distinct employee names.

```
mysql> select distinct name from employee;
+----------+
| name     |
+----------+
| Aarav    |
| Karan    |
| Arun     |
| Mira     |
| Adrian   |
| Varun    |
| Nina     |
| Aman     |
| Rohan    |
| Pranav   |
| Meera    |
| Aarhan   |
| Ira      |
| Arav     |
| Kiran    |
| Mohan    |
| Sarah    |
| Niranjan |
| Asha     |
| Dev      |
| Anand    |
| Rita     |
| Brian    |
+----------+
23 rows in set (0.00 sec)
```

20. Write a query to count the number of departments having the same location.

```
mysql> select city, count(dept_id) as same_city_department from department
    -> group by city;
+--------+----------------------+
| city   | same_city_department |
+--------+----------------------+
| Mumbai |                    2 |
| Pune   |                    1 |
| Delhi  |                    1 |
| Nagpur |                    1 |
+--------+----------------------+
4 rows in set (0.00 sec)
```